Project Title

Deploying a Web Application on AWS EC2 Using Apache Web Server

Project Objective

The goal of this project is to help gain hands-on experience with **Amazon Web Services (AWS)** by launching an **EC2 instance**, configuring **SSH and HTTP access**, installing the **Apache2 web server**, and deploying a **sample web application** accessible through a browser.

By the end of this project, students will understand basic cloud infrastructure setup, Linux server management, and web application deployment.

Prerequisites

- Basic knowledge of Linux commands
- An active AWS account
- Basic understanding of networking (ports, security groups)
- SSH client (Terminal on Linux/macOS or PuTTY on Windows)

STEP 1: Create an EC2 Instance

1. Login to AWS Console

- Open AWS Management Console → Go to **EC2 Service** → Click **Launch Instance**

2. Choose AMI & Instance Type

- Select **Ubuntu Server 20.04 / 22.04 LTS**
- Choose **t2.micro (Free Tier)**

3.**Create Key Pair**

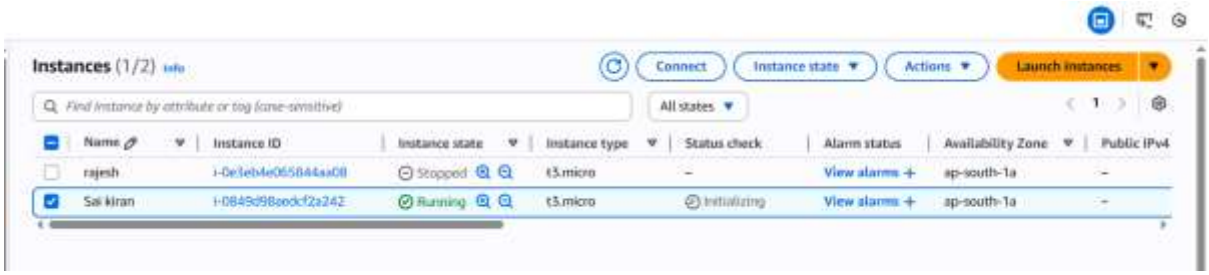
- Create or select existing **Key Pair**
- Download .pem file (used for SSH login)

4. **Configure Network & Storage**

- Select **VPC & Public Subnet**
- Enable **Public IP**
- Allow **SSH (Port 22)**
- Keep default **8 GB storage**

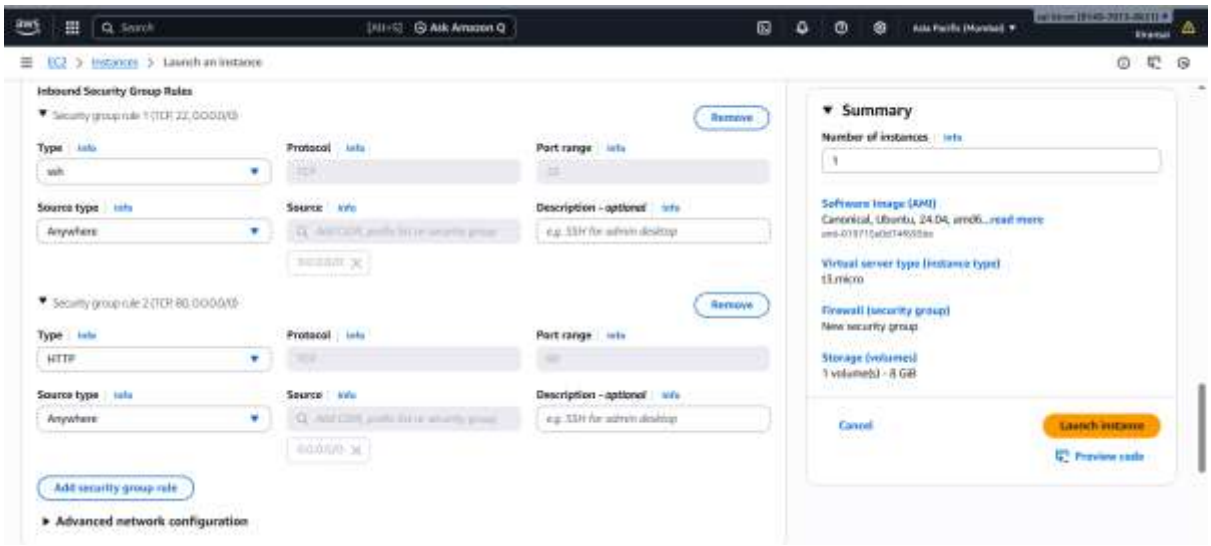
5.**Launch & Connect Instance**

- Click **Launch Instance**
- Connect using SSH: ssh -i key.pem ubuntu@public-ip



STEP 2 : Configure Security Groups (Firewall Rules)

1. **Login to AWS Console**
   → Open **EC2 Dashboard**
2. **Go to Security Groups**
   → In the left panel, click **Security Groups**
3. **Select Your Security Group**
   → Choose the security group attached to your EC2 instance
4. **Edit Inbound Rules**
   → Click **Edit inbound rules**
5. **Add SSH Rule (For Server Access)**
   - Type: **SSH**
   - Protocol: **TCP**
   - Port: **22**
   - Source: **My IP** (Allows only your system to connect securely)
6. **Add HTTP Rule (For Website Access)**
   - Type: **HTTP**
   - Protocol: **TCP**
   - Port: **80**
   - Source: **Anywhere (0.0.0.0/0)** (Allows public users to access the website)
7. **Save Rules**
   → Click **Save rules**



Step 3: Connect to EC2 Using SSH

1. **Open AWS EC2 Dashboard**
   → Go to **Instances**
   → Select your running EC2 instance
2. **Copy Public IP Address**
   → Find **Public IPv4 address**
   → Copy the IP address
3. **Open Terminal / Command Prompt**
   - Linux / Mac → Open **Terminal**
   - Windows → Use **Git Bash** or **PowerShell**
4. **Go to Key Pair Location**
   → Navigate to the folder where your .**pem** key file is stored
5. **Connect Using SSH Command**
6. ssh -i your-key.pem ubuntu@<public-ip>

   Replace:

   - your-key.pem with your key file name
   - <public-ip> with your EC2 public IP address
7. **Verify Connection**
   → If login is successful, you will see the Ubuntu server terminal
   → You can now run Linux commands on your EC2 instance

```
/usr/bin/xauth:  file /home/ubuntu/.Xauthority does not exist
To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

ubuntu@ip-10-0-60-184:~$
ubuntu@ip-10-0-60-184:~$
ubuntu@ip-10-0-60-184:~$
ubuntu@ip-10-0-60-184:~$
ubuntu@ip-10-0-60-184:~$
```

Step 4: **Install and Start Apache2 Web Server**

1. **Update System Packages**
   → Run the following command to update package information:

   Command: sudo apt update

2. **Install Apache2**
   → Install the Apache web server:

   Command: sudo apt install apache2 -y

3. **Start Apache Service**
   → Start the Apache server:

   Command: sudo systemctl start apache2

4. **Enable Apache at Boot**
   → Make Apache start automatically when the server restarts:

   Command: sudo systemctl enable apache2

5. **Check Apache Status**
   → Verify Apache is running properly:

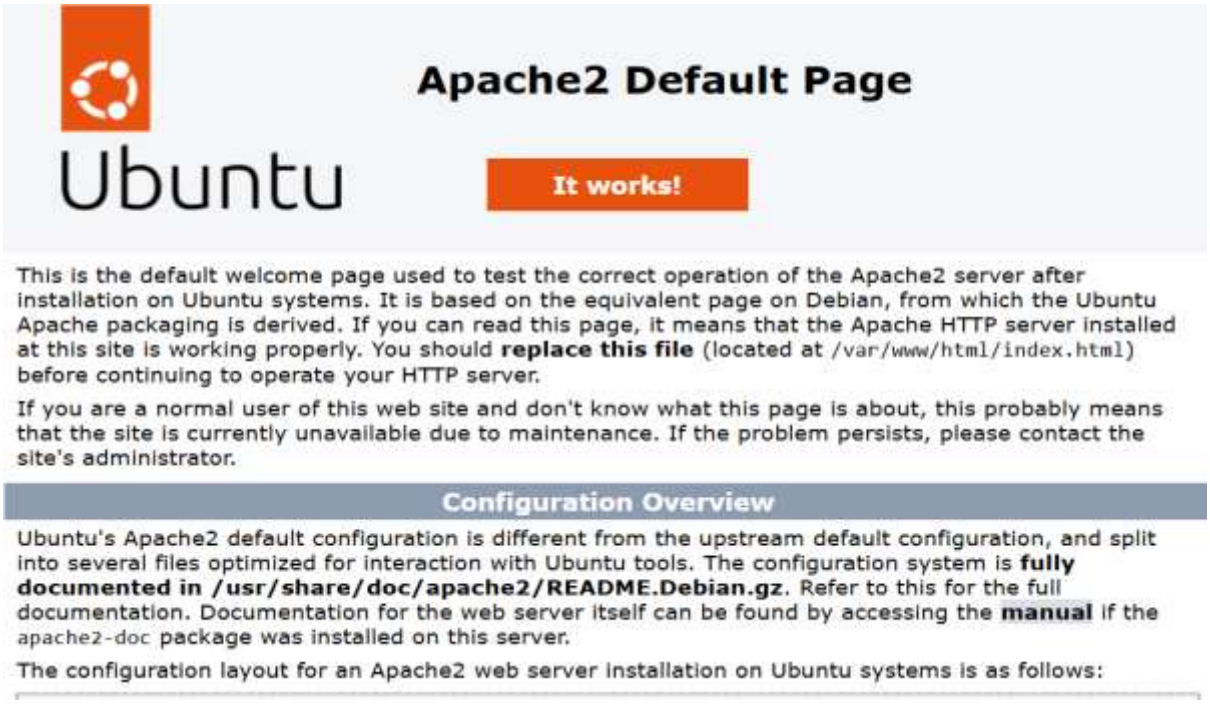
   Command: sudo systemctl status apache2

```
ubuntu@ip-10-0-60-184:~$ sudo systemctl status apache2
● apache2.service - The Apache HTTP Server
     Loaded: loaded (/usr/lib/systemd/system/apache2.service; enabled; preset: enabled)
     Active: active (running) since Wed 2026-02-04 08:51:35 UTC; 1min 59s ago
       Docs: https://httpd.apache.org/docs/2.4/
   Main PID: 2189 (apache2)
      Tasks: 55 (limit: 1008)
     Memory: 5.3M (peak: 5.7M)
        CPU: 46ms
     CGroup: /system.slice/apache2.service
             ├─2189 /usr/sbin/apache2 -k start
             ├─2192 /usr/sbin/apache2 -k start
             └─2193 /usr/sbin/apache2 -k start

Feb 04 08:51:35 ip-10-0-60-184 systemd[1]: Starting apache2.service - The Apache HTTP Server...
Feb 04 08:51:35 ip-10-0-60-184 apachectl[2188]: AH00558: apache2: Could not reliably determine the server's fully qu
Feb 04 08:51:35 ip-10-0-60-184 systemd[1]: Started apache2.service - The Apache HTTP Server.
```

STEP5: Test HTTP Access to Apache Web Server

1. **Open a Web Browser**
   → Use any browser (Chrome, Edge, Firefox, etc.)
2. **Enter EC2 Public IP Address**
   → Type the EC2 **Public IPv4 address** in the address bar
3. http://<public-ip>
4. **Verify the Apache Page**
   → If Apache is working, the **Apache2 Ubuntu Default Page** will be displayed



STEP 6: Deploy a Sample Web Application

1. Go to Apache Web Root Directory
→ Run the command: cd /var/www/html

2. Remove Default Apache Page (Optional)
→ Delete the default index file: sudo rm index.html

3. Create or Upload Your Web Application File
→ Example: Create a simple HTML page

```
sudo nano index.html
```

→ Save and exit (Press **CTRL + X → Y → Enter**)

4. Refresh the Browser
→ Open browser and enter: http://<public-ip>

5.Verify Deployment
→ Your sample web application page should be displayed